



KTU NOTES

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NOTIFICATIONS | SOLVED QUESTION PAPER**

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Characteristic equation is $1 + G(s)H(s) = 0$

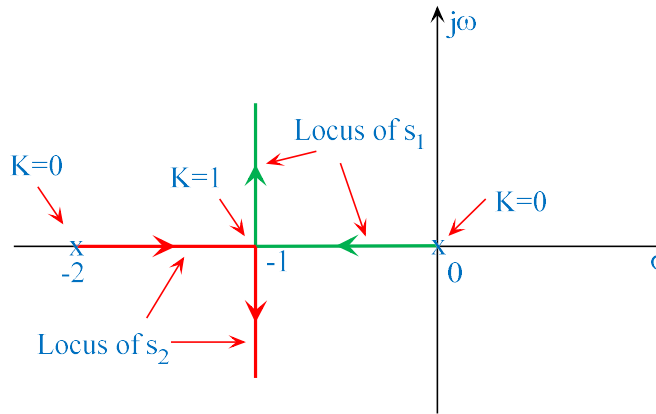
$$s^2 + 2s + K = 0$$

Roots of characteristic equation, $s_1, s_2 = -1 \pm \sqrt{1-K}$

$$s_1 = -1 + \sqrt{1-K} \text{ and } s_2 = -1 - \sqrt{1-K}$$

$K = 0$	$s_1 = 0$	$s_2 = -2$	Originates
$0 < K < 1$	Real and negative	Real and negative	
$K = 1$	$s_1 = -1$	$s_2 = -1$	Break-away
$K > 1$	Complex with real part -1	Complex conjugate with real part -1	
$K = \infty$			Terminate

The path taken by the roots of the characteristic equation when open from 0 to ∞ (or any other system parameter is varied) is called root locus.



Here, root locus originates from $s_1 = 0$ and $s_2 = -2$ and moves along negative real axis. 1. The two root loci meet at -1 and then proceeds to infinity at angle $\pm 90^\circ$ to the real axis.

$$R(s) = 1 + G(s)H(s)$$

Characteristic equation is given by,

$$1 + G(s)H(s) = 0$$

$$G(s)H(s) = -1 = 1 \angle \pm 180^\circ \quad (\text{angle shall be odd multiple of } 180^\circ)$$

$$|G(s)H(s)| = 1 \text{ ----- magnitude criterion}$$

$$\angle G(s)H(s) = \pm (2q+1)180^\circ; q = 0, 1, 2, \text{----- angle criterion}$$

	The values of s which fulfill both the angle and magnitude criteria are the roots of the characteristic equation or the closed-loop poles.
	We first apply the angle criterion to determine whether a point in the s-plane is on the root locus. If this condition is satisfied, then we can use the magnitude criterion to determine the value of K for which this will be a root of the characteristic equation.

The open loop transfer function is $G(s) = \frac{K}{s(s+4)(s+5)}$. Check whether s = -1 is on the root locus.

$$G(s)H(s) \Big|_{s=-1} = \frac{K}{-1 \times 3 \times 4} = \frac{-K}{12} = \frac{K}{12} \angle 180^\circ$$

$\angle G(s)H(s) = 180^\circ$ Angle criterion is satisfied; hence s = -1 is on root locus.

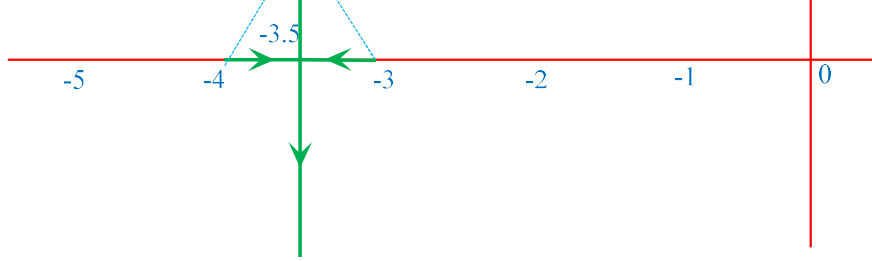
Magnitude criterion, $|G(s)H(s)| = 1$

$$\frac{K}{12} = 1 \quad K = 12.$$

For K = 12, one of the roots is located at s = -1.

The open loop transfer function is $G(s) = \frac{K}{s(s+2)}$. Check whether s = -1 + j is on the root locus.

2	Each branch of the root locus originates from an open loop pole corresponding to $K = 0$. Each branch terminates at either open loop zero or infinity corresponding to $K = \infty$. Number of branches terminating at infinity equals to the difference between the number of poles and number of zeros of $G(s)H(s)$.
3	A point on the real axis lies on the root locus if the sum of the poles to the right of the point is an odd number.
4	The $(n-m)$ root locus branches that proceed to infinity do so along the asymptotes. $\phi_A = \frac{\pm 180(2q + 1)}{n - m} \quad q = 0, 1, 2, \dots, (n - m - 1)$ n = number of open loop poles and m = number of open loop zeros
5	The centroid, the point of intersection of the asymptotes with real axis is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n - m}$
6	The breakaway points (points at which multiple roots of the characteristic equation occur) on the root locus are the solution of $\frac{dK}{ds} = 0$
7	The angle of departure/arrival (if RL leaves from a complex pole/arrives at a complex zero) is given by The angle of departure at open loop pole is given by $\phi_d = 180^\circ - \phi$ where ϕ is net angle contribution to this pole by all other poles and zeros $\phi = \sum \phi_p - \sum \phi_z$ $\sum \phi_p$ = sum of angles subtended by the phasors drawn to this pole from all other poles $\sum \phi_z$ = sum of angles subtended by the phasors drawn to this pole from all other zeros The angle of arrival at open loop zero is given by $\phi_a = 180^\circ + \phi$ where ϕ is net angle contribution to the zero under consideration from all other poles and zeros. $\phi = \sum \phi_p - \sum \phi_z$ $\sum \phi_p$ = sum of angles subtended by the phasors drawn to this zero from all other poles $\sum \phi_z$ = sum of angles subtended by the phasors drawn to this zero from all other zeros



$$\text{Net angle contribution at } P = \angle G(s) = \angle \frac{K}{(-3.5 + j1 + 3)(-3.5 + j1 + 4)} = 180^\circ$$

$$\text{Net angle contribution at } P_1 = \angle G(s) = \angle \frac{K}{(-5 + j1 + 3)(-5 + j1 + 4)} = 71.6^\circ$$

$$\text{Net angle contribution at } P_1 = \angle G(s) = \angle \frac{K}{(-2 + j1 + 3)(-2 + j1 + 4)} = -71.6^\circ$$

Net angle contribution on root-locus is always 180° . Net angle contribution towards positive and towards right is negative.

Plot the root locus of a system whose forward path transfer function is $G(s) =$

1	Number of branches of root locus = order of the characteristic polynomial
2	Branches of the root locus originates from open loop poles at $s = -3$ and $s = -3.5$ and goes to infinity.
3	On the real axis, the root locus is found between section $s = -3$ and $s = -3.5$. The number of poles and zeros to the right of this section is odd.
4	2 branches proceed to infinity along the asymptotes $\phi_A = \frac{\pm 180(2q + 1)}{n - m} \quad q = 0, 1$ $q = 0 \Rightarrow \phi_A = \pm 90^\circ$ $q = 1 \Rightarrow \phi_A = \pm 270^\circ$

Substitute $s = j\omega$,

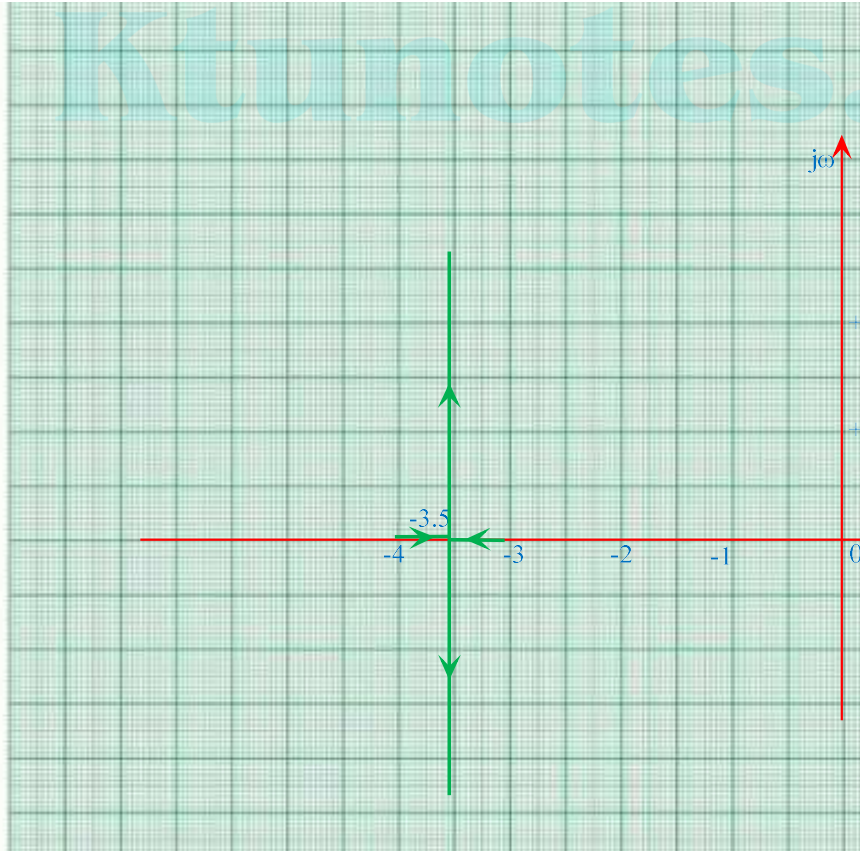
$$-\omega^2 + j7\omega + 12 + K = 0$$

Equating real and imaginary parts,

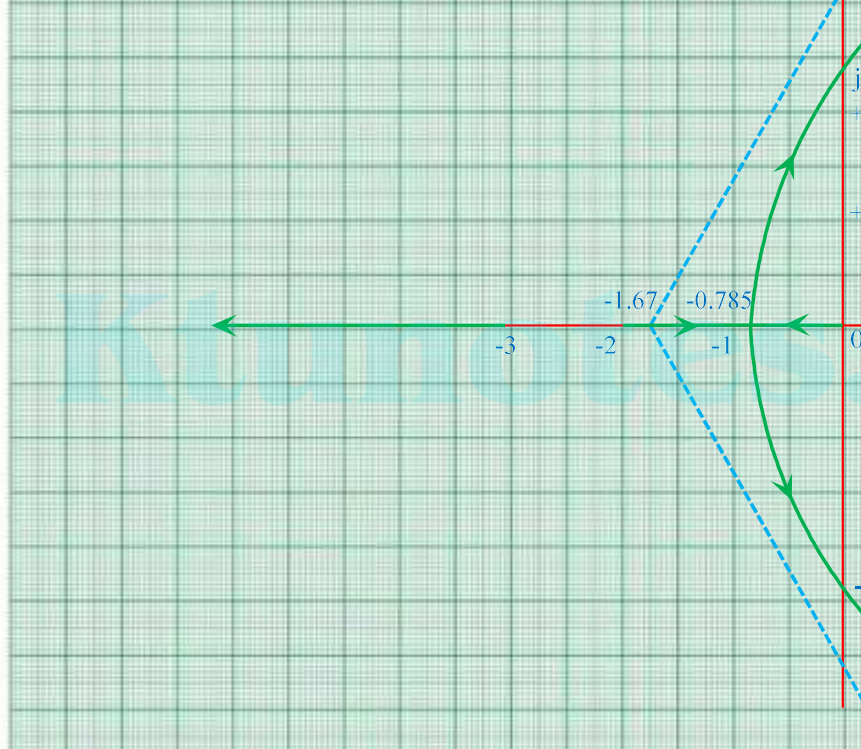
$$7\omega = 0 \quad \text{and} \quad -\omega^2 + 12 + K = 0$$

$$\omega = 0 \quad \text{and} \quad K = -12 ; K \text{ is negative}$$

No intersection with imaginary axis.



	$q = 1 \Rightarrow \phi_A = \pm 180^\circ$ $q = 2 \Rightarrow \phi_A = \pm 300^\circ$ (same as 60°)
5	The centroid (intersection of asymptotes) is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n - m} = \frac{0 - 2 - 3}{3} = -1.67$
6	Breakaway points $\frac{dK}{ds} = 0$ $s^3 + 5s^2 + 6s + K = 0$ $K = -s^3 - 5s^2 - 6s$ $\frac{dK}{ds} = -3s^2 - 10s - 6 = 0$ $3s^2 + 10s + 6 = 0$ $s = -0.785 \text{ or } -2.549 \text{ (But, -2.549 does not lie on root locus)}$ $s = -0.785$
7	Intersection with imaginary axis $s^3 + 5s^2 + 6s + K = 0$ $\begin{array}{r l} s^3 & 1 & 6 \\ s^2 & 5 & K \\ s^1 & 30 - K & \\ s^0 & 5 & \\ & K & \end{array}$ $K > 0$ $30 - K > 0$ $K < 30$ $0 < K < 30$ For $K = 30$, all elements in one row equal zeros $s^3 + 5s^2 + 6s + 30 = 0$ Auxiliary equation, $5s^2 + 30 = 0$



Plot the root locus of a system whose forward path transfer function is $G(s) =$

1	Number of branches of root locus = order of the characteristic polynomial.
2	Branches of the root locus originates from open loop poles at $s = 0$ at $-j3$ and all end at infinity.
3	The entire negative real axis is part of the root locus because the sum of zeros to the right of this section is odd.

$$\frac{dK}{ds} = -3s^2 - 8s - 13 = 0$$

$$3s^2 + 8s + 13 = 0$$

$$s = -1.33 \pm j1.16$$

Note: Breakaway point may be real or complex. In order to determine point, substitute the breakaway point value in the equation of K to get value of K is real positive, the breakaway point is valid one.

$$K = -s^3 - 4s^2 - 13s$$

When $s = -1.33 + j1.16$,

$$K = -(-1.33 + j1.16)^3 - 4(-1.33 + j1.16)^2 - 13(-1.33 + j1.16) = 12$$

When $s = -1.33 - j1.16$,

$$K = -(-1.33 - j1.16)^3 - 4(-1.33 - j1.16)^2 - 13(-1.33 - j1.16) = 12$$

Hence, $s = -1.33 \pm j1.16$ is not a valid breakaway point.

7

The angle of departure at open loop pole $(-2+j3)$ is given by

$$\varphi_d = 180 - \phi$$

Net angle contribution to this pole,

$$\phi = \sum \phi_p - \sum \phi_z = \angle_0^{-2+j3} + \angle_{-2-j3}^{-2+j3} = 123.7 + 90 = 213.7$$

$$\varphi_d = 180 - 213.7 = -33.7^\circ$$

The angle of departure at open loop pole $(-2-j3)$ is given by

$$\varphi_d = 180 - \phi$$

Net angle contribution to this pole,

$$\phi = \sum \phi_p - \sum \phi_z = \angle_0^{-2-j3} + \angle_{-2+j3}^{-2-j3} = -123.7 - 90 = -213.7$$

$$\varphi_d = 180 + 213.7 = 393.7^\circ = +33.7^\circ$$

8

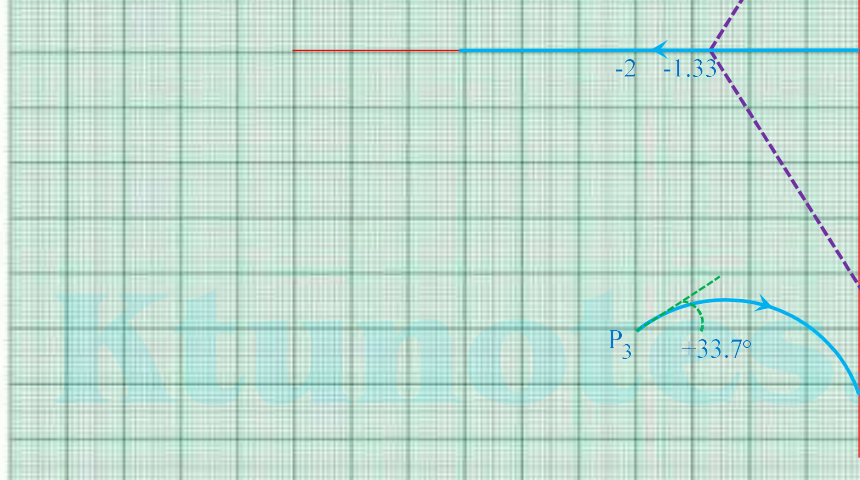
Intersection with imaginary axis

$$s^3 + 4s^2 + 13s + K = 0$$

Substitute $s = j\omega$,

$$-j\omega^3 - 4\omega^2 + j13\omega + K = 0$$

Equating real and imaginary parts,



Plot the root locus of a system whose forward path transfer function is

$$G(s) = \frac{K}{s(s+4)(s^2+4s+20)}.$$

1	Number of branches of root locus = order of the characteristic polynomial
2	Branches of the root locus originates from open loop poles at $s = 0$, $s = -2 - j4$ and all end at infinity.
3	On the real axis, the root locus is found between $s = 0$ and $s = -4$ because the number of poles and zeros to the right of this section is odd.
4	4 branches proceed to infinity along the asymptotes $\phi_A = \frac{\pm 180(2q+1)}{n-m} \quad q = 0, 1, 2, 3$ $q = 0 \Rightarrow \phi_A = \pm 45^\circ$ $q = 1 \Rightarrow \phi_A = \pm 135^\circ$ $q = 2 \Rightarrow \phi_A = \pm 225^\circ \text{ (same as } \pm 135^\circ)$ $q = 3 \Rightarrow \phi_A = \pm 315^\circ \text{ (same as } \pm 45^\circ)$
5	The centroid (intersection of asymptotes) is given by

value of K is real positive, the breakaway point is valid one.

$$K = -s^4 - 8s^3 - 36s^2 - 80s$$

For $s_1 = -2$,

$$K = -(-2)^4 - 8(-2)^3 - 36(-2)^2 - 80(-2) = 64$$

For $s_2 = -2 + j2.45$,

$$K = -(-2 + j2.45)^4 - 8(-2 + j2.45)^3 - 36(-2 + j2.45)^2 - 80(-2 + j2.45)$$

For $s_3 = -2 - j2.45$,

$$K = -(-2 - j2.45)^4 - 8(-2 - j2.45)^3 - 36(-2 - j2.45)^2 - 80(-2 - j2.45)$$

Hence, $s_1 = -2$; $s_2 = -2 + j2.45$; $s_3 = -2 - j2.45$ are valid breakaway

7

The angle of departure at open loop pole $(-2+j4)$ is given by

$$\varphi_d = 180 - \phi$$

Net angle contribution to this pole,

$$\phi = \sum \phi_p - \sum \phi_z = \angle_0^{-2+j4} + \angle_{-4}^{-2+j4} + \angle_{-2-j4}^{-2+j4} = 116.57 + 63.43 + 90$$

$$\varphi_d = 180 - 270 = -90^\circ$$

The angle of departure at open loop pole $(-2-j3)$ is given by

$$\varphi_d = 180 - \phi$$

Net angle contribution to this pole,

$$\phi = \sum \phi_p - \sum \phi_z = \angle_0^{-2-j4} + \angle_{-4}^{-2-j4} + \angle_{-2+j4}^{-2-j4} = -116.57 - 63.43 -$$

$$\varphi_d = 180 + 270 = 450^\circ = +90^\circ$$

8

Intersection with imaginary axis

$$s^4 + 8s^3 + 36s^2 + 80s + K = 0$$

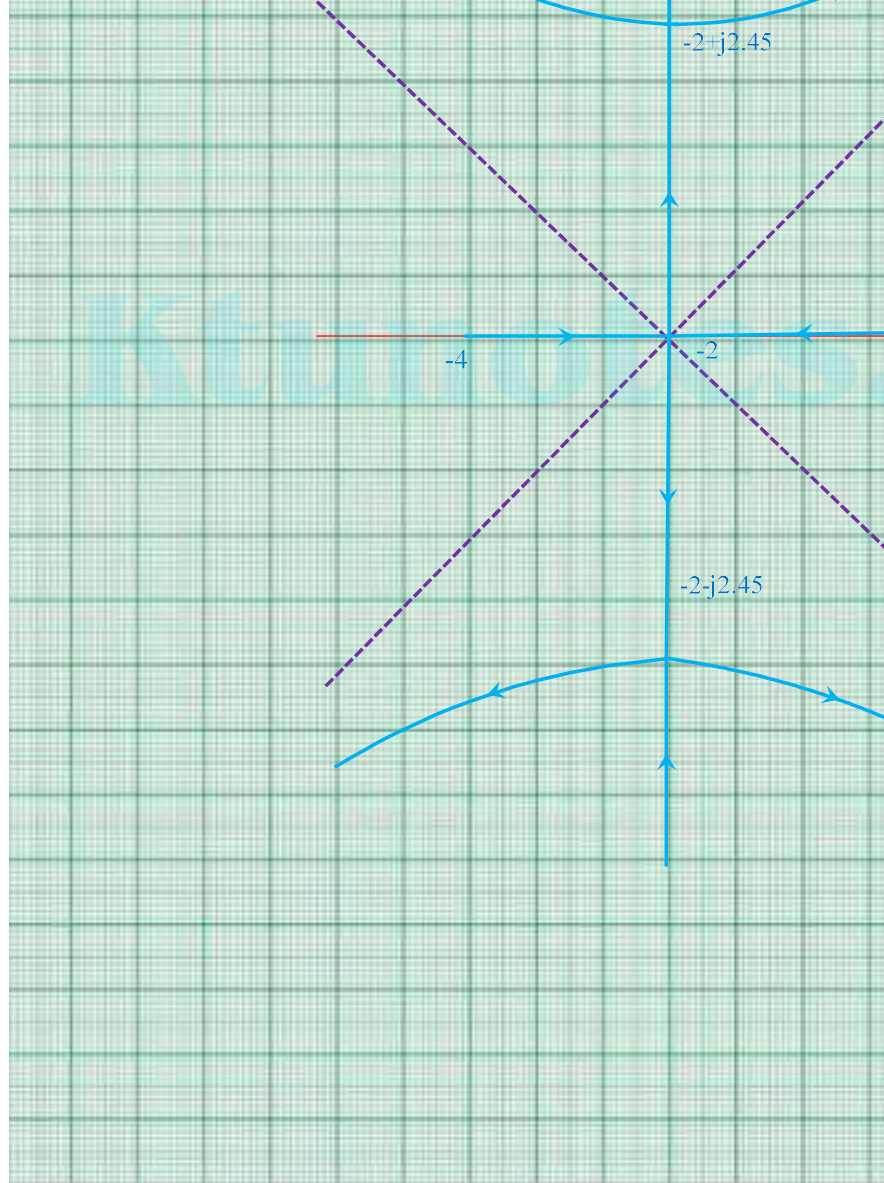
Substitute $s = j\omega$,

$$\omega^4 - j8\omega^3 - 36\omega^2 + j80\omega + K = 0$$

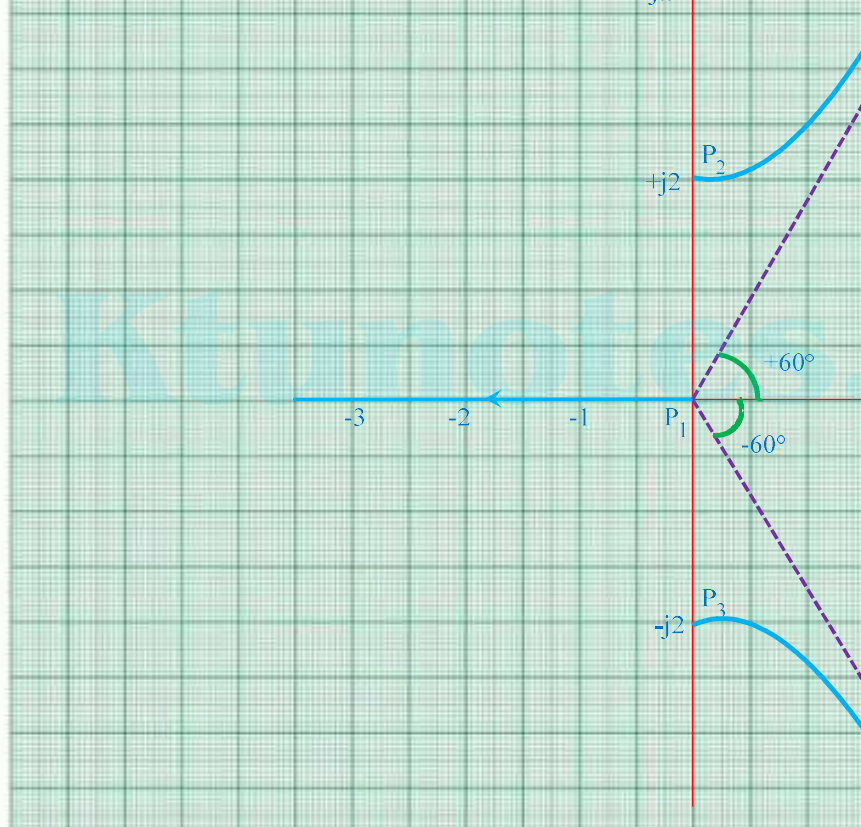
Equating real and imaginary parts,

$$-8\omega^3 + 80\omega = 0 \quad \text{and} \quad \omega^4 - 36\omega^2 + K = 0$$

$$\omega = \pm 3.16 \quad \text{and} \quad K = 260$$



	$q = 2 \Rightarrow \phi_A = \pm 300^\circ$ (same as 60°)
5	The centroid (intersection of asymptotes) is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n - m} = \frac{0 + j3 - j3}{3} = 0$
6	Breakaway points $\frac{dK}{ds} = 0$ Characteristic equation is $s^3 + 4s + K = 0$ $K = -s^3 - 4s$ $\frac{dK}{ds} = -3s^2 - 4 = 0$ $3s^2 + 4 = 0$ $s = \pm j1.15$ Note: Breakaway point may be real or complex. In order to determine the breakaway point, substitute the breakaway point value in the equation of K to get the value of K. If the value of K is real and positive, the breakaway point is a valid one. $K = -s^3 - 4s$ When $s = +j1.15$, $K = -(j1.155)^3 - 4(j1.155) = -j3.08$ When $s = -j1.155$, $K = +j3.08 \text{ (complex conjugate)}$ Hence, $s = \pm j1.155$ is not a valid breakaway point.
7	The angle of departure at open loop pole (+j2) is given by $\varphi_d = 180 - \phi$ Net angle contribution to this pole, $\phi = \sum \phi_p - \sum \phi_z = \angle_0^{+j2} + \angle_{-j2}^{+j2} = 90 + 90 = 180^\circ$ $\varphi_d = 180 - 180 = 0^\circ$ The angle of departure at open loop pole (-j2) is given by $\varphi_d = 180 - \phi$ Net angle contribution to this pole,



Plot the root locus of a system whose forward path transfer function is $G(s) =$

1	Number of branches of root locus = order of the characteristic polynomial
2	Branches of the root locus originates from open loop poles at $s = 0$, at one ends at zero $s = -1$ and two end at infinity.

$$K = \frac{-s^3 - 3.6s^2}{s+1} = -\frac{s^3 + 3.6s^2}{s+1}$$

$$\frac{dK}{ds} = \frac{(s+1)(3s^2 + 7.2s) - (s^3 + 3.6s^2)}{(s+1)^2} = 0$$

$$(s+1)(3s^2 + 7.2s) - (s^3 + 3.6s^2) = 0$$

$$3s^3 + 7.2s^2 + 3s^2 + 7.2s - s^3 - 3.6s^2 = 0$$

$$2s^3 + 6.6s^2 + 7.2s = 0$$

$$s^3 + 3.3s^2 + 3.6s = 0$$

$$s = 0 ; s = -1.65 \pm j0.936$$

Note: Breakaway point may be real or complex. In order to determine the breakaway point, substitute the breakaway point value in the equation of K to get the value of K. If the value of K is real and positive, the breakaway point is a valid one.

$$K = -\frac{s^3 + 3.6s^2}{s+1}$$

When $s = 0$,

$$K = 0;$$

$s = 0$ is a valid breakaway point

When $s = 1.65 + j0.936$,

$$K = 6.83 \angle 49.785^\circ$$

Hence, $s = 1.65 \pm j0.936$ are not valid breakaway points.

7

Since there are no complex poles or complex zeros, there is no need for departure or arrival.

8

Intersection with imaginary axis

$$s^3 + 3.6s^2 + Ks + K = 0$$

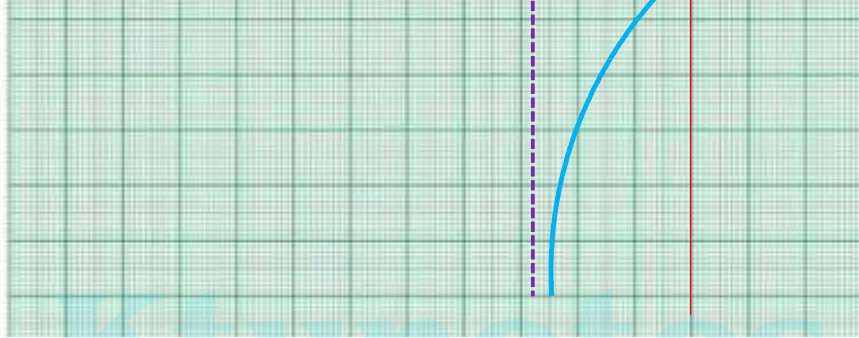
Substitute $s = j\omega$,

$$-j\omega^3 - 3.6\omega^2 + jK\omega + K = 0$$

Equating real and imaginary parts,

$$-\omega^3 + K\omega = 0 \quad \text{and} \quad -3.6\omega^2 + K = 0$$

$$\omega = 0 \quad \text{and} \quad K = 0$$



Plot the root locus of a system whose forward path transfer function is

$$G(s) = \frac{K}{s(s+6)(s^2+4s+13)}. \text{ Also, comment on the stability of the system.}$$

1	Number of branches of root locus = order of the characteristic polynomial
2	Branches of the root locus originates from open loop poles at $s = 0$, $s = -6$ and $s = -2 - j3$ and all end at infinity.
3	On the real axis, the root locus is found between $s = 0$ and $s = -6$ because the number of poles and zeros to the right of this section is odd.
4	4 branches proceed to infinity along the asymptotes $\phi_A = \frac{\pm 180(2q+1)}{n-m} \quad q = 0, 1, 2, 3$ $q = 0 \Rightarrow \phi_A = \pm 45^\circ$ $q = 1 \Rightarrow \phi_A = \pm 135^\circ$ $q = 2 \Rightarrow \phi_A = \pm 225^\circ \text{ (same as } \pm 135^\circ)$ $q = 3 \Rightarrow \phi_A = \pm 315^\circ \text{ (same as } \pm 45^\circ)$
5	The centroid (intersection of asymptotes) is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n-m} = \frac{0 - 6 - 2 + j3 - 2 - j3}{4} = -2.5$

	$K = -(-4.2)^4 - 10(-4.2)^3 - 37(-4.2)^2 - 78(-4.2) = 104.6$ For $s_2 = -1.65 + j1.386$, $K = -(-1.65 + j1.386)^4 - 10(-1.65 + j1.386)^3 - 37(-1.65 + j1.386)^2 - 78(-1.65 + j1.386) - 78 = 69.15 - j18.12$ For $s_3 = -1.65 - j1.386$, $K = 69.15 + j18.12 \text{ (complex conjugate)}$ Hence, only $s_1 = -4.2$ is valid breakaway point.
7	The angle of departure at open loop pole $(-2+j3)$ is given by $\varphi_d = 180 - \phi$ Net angle contribution to this pole, $\phi = \sum \phi_p - \sum \phi_z = \angle_0^{-2+j3} + \angle_{-6}^{-2+j3} + \angle_{-2-j3}^{-2+j3} = 123.7 + 36.87 + 90 = 250.57^\circ$ $\varphi_d = 180 - 250.57 = -70.57^\circ$ The angle of departure at open loop pole $(-2-j3)$ is given by $\varphi_d = 180 - \phi$ Net angle contribution to this pole, $\phi = \sum \phi_p - \sum \phi_z = \angle_0^{-2-j4} + \angle_{-4}^{-2-j4} + \angle_{-2+j4}^{-2-j4} = -123.7 - 36.87 - 90 = -250.57^\circ$ $\varphi_d = 180 + 250.57 = 430.57^\circ = +70.57^\circ$
8	Intersection with imaginary axis $s^4 + 10s^3 + 37s^2 + 78s + K = 0$ Substitute $s = j\omega$, $\omega^4 - j10\omega^3 - 37\omega^2 + j78\omega + K = 0$ Equating real and imaginary parts, $-10\omega^3 + 78\omega = 0 \quad \text{and} \quad \omega^4 - 37\omega^2 + K = 0$ $\omega = \pm 2.8 \text{ and } K = 228$

For $0 < K < 228$, the system is stable because roots lie on the left-hand side of s -plane.

For $K = 228$, the system is marginally stable because roots lie on the imaginary axis.

For $K > 228$, the system is unstable because roots lie on the right-hand side of s -plane.



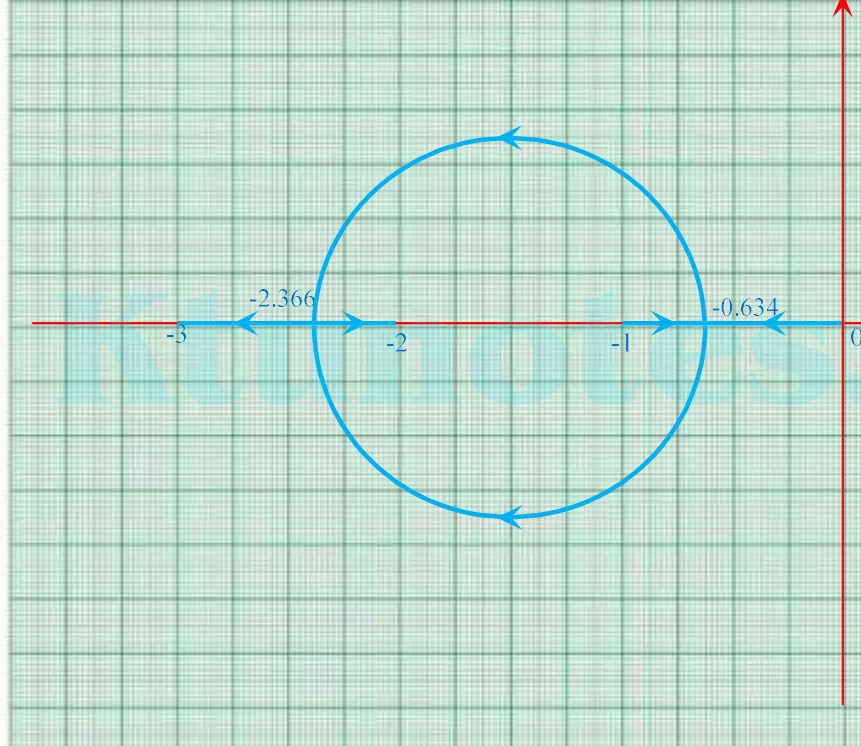
Plot the root locus of a system whose forward path transfer function is $G(s) =$

1	Number of branches of root locus = order of the characteristic polynomial
2	Branches of the root locus originates from open loop poles at $s = 0$ and $s = -3$ and open loop zeros at $s = -3$ and $s = -\infty$.
3	On the real axis, the root locus is found (a) between section $s = 0$ and $s = -3$ and section $s = -3$ and $s = -\infty$ because the sum of number of poles and zeros in each section is odd.
4	1 branch proceed to infinity along the asymptotes with angles $\phi_A = \frac{\pm 180(2q + 1)}{n - m} \quad q = 0 \text{ (here } n - m - 1 = 0)$ $q = 0 \Rightarrow \phi_A = \pm 180^\circ$
5	The centroid (intersection of asymptotes) is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n - m} = \frac{0 - 2 + 3}{2 - 1} = 1$
6	Breakaway points

	$s = -1.268$ or -4.732 (Both lie on root locus; hence they are break-away points) (since $s = -4.732$ lies between two zeros, it is a break-in point)
7	$s^2 + 2s + K(s + 3) = 0$ $s^2 + (2 + K)s + 3K = 0$ Substitute $s = j\omega$, $-\omega^2 + j\omega(2 + K) + 3K = 0$ Equating real and imaginary parts, $\omega = 0$ and $K = -2$ (not possible) The root locus does not cross the imaginary axis.

Plot the root locus of a system whose forward path transfer function is $G(s) =$

1	Number of branches of root locus = order of the characteristic polynomial
2	Branches of the root locus originates from open loop poles at $s = 0$ and $s = -1$ and open loop zeros at $s = -2$ and $s = -3$.
3	On the real axis, the root locus is found (a) between section $s = 0$ and $s = -1$ and (b) between section $s = -2$ and $s = -3$ because the sum of number of poles and zeros in each section is odd.
4	Breakaway points $\frac{dK}{ds} = 0$ Characteristic equation is $1 + \frac{K(s+2)(s+3)}{s(s+1)} = 0$ $K = -\frac{s^2 + s}{s^2 + 5s + 6}$ $\frac{dK}{ds} = -(s^2 + 5s + 6)(2s + 1) + (s^2 + s)(2s + 5) = 0$ $4s^2 + 12s + 6 = 0$ $2s^2 + 6s + 3 = 0$



EFFECT OF ADDITION OF POLES

1	Shape of root locus changes and it shifts towards the imaginary axis
2	Relative stability decreases
3	System becomes more oscillatory in nature
4	Range of K for stability decreases
5	Settling time increases

DESIGN OF COMPENSATORS USING ROOT LOCUS

The root locus approach is used for design purpose when the specification is given. The two main specifications which are used in the root locus compensation are the undamped natural frequency ω_n . The dominant closed-loop poles near $j\omega$ -axis are the specifications. Therefore, the compensation using root locus reshapes the root locus, so that the dominant closed-loop poles at the desired locations are placed. This is done by root locus compensation using lead, lag or lag-lead network.

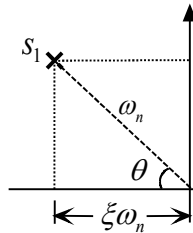
$$\text{Complex pole, } s_1 = -\xi\omega_n + j\omega_n\sqrt{1-\xi^2}$$

$$\theta = \cos^{-1} \xi$$

$$\text{Peak overshoot, } M_p = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}$$

$$\text{Settling time, } t_s = \frac{4}{\xi\omega_n}$$

$$\text{Rise time, } t_r = \frac{\pi - \theta}{\omega_n\sqrt{1-\xi^2}}$$



What is the best damping ratio to use? Selection of ξ of industrial control applications is a trade-off between relative stability & speed of response. A smaller ξ decreases normalised peak overshoot but increases $\%M_p$. Practically, ξ is between 0.4 & 0.7 (ie. M_p is between 5 to 20%). For accurate system design, ξ should be small as possible to reduce steady state error, which is proportional to ξ .

that the compensated system has a root locus that goes through the desired pole of gain. One of the advantages of compensating a system in this way is that additional poles can be added at the low-power end of the system before the plant. Addition of compensating zeros need not interfere with the power output requirements of the system or present other design problems. The compensating poles and zeros can be generated with a passive network.

A possible disadvantage of compensating a system with additional open-loop poles is that the system order can increase, with a subsequent effect on the desired response.

Note: Adding only zero is often problematic because such controller amplifies the high-frequency noise. Adding only pole is often problematic because such controller generates a less stable system (i.e., moves the closed loop poles to the right).

LEAD COMPENSATOR

For lead compensator, $G_c(s) = K \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}$ where $0 < \alpha < 1$ and lag compensator

where $\beta > 1$.

Both a lead compensator and a lag compensator have the same shape: So they

and a pole at $s = \frac{-1}{\alpha T}$ or $s = \frac{-1}{\beta T}$. For lead compensators the pole lies more to the left of the zero

plane than the zero and vice versa for lag compensators

RULES FOR DESIGN OF LEAD COMPENSATOR USING ROOT LOCUS	
1	List the system specifications and translate them into a desired root locus.
2	Sketch the uncompensated root locus, and determine whether the desired root location can be realized with an uncompensated system (with gain adjustment only).
3	If a compensator is necessary, place the zero of the phase-lead network at the desired root location or to the left of the first two real poles.

2	Branches of the root locus originates from open loop poles at $s = 0$ and at infinity.
3	On the real axis, the root locus is found between section $s = 0$ and $s =$
4	2 branches proceed to infinity along the asymptotes $\phi_A = \frac{\pm 180(2q + 1)}{n - m} \quad q = 0, 1$ $q = 0 \Rightarrow \phi_A = \pm 90$ $q = 1 \Rightarrow \phi_A = \pm 270$
5	The centroid (intersection of asymptotes) is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n - m} = \frac{0 - 8}{2} = -4$
6	Breakaway points $\frac{dK}{ds} = 0$ $s^2 + 8s + K = 0$ $K = -s^2 - 8s$ $\frac{dK}{ds} = -2s - 8 = 0$ $s = -4$
7	Since there is no complex poles and zeros, no angle of departure or arrival with imaginary axis.

1	DOMINANT POLE: $M_p = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}$ $\frac{-\xi\pi}{\sqrt{1-\xi^2}} = \ln(M_p) = -2.354$ $\xi\pi = 2.354 \times \sqrt{1-\xi^2}$ $(\xi\pi)^2 = 2.354^2 \times (1-\xi^2)$ $\xi = \frac{2.354}{\sqrt{2.354^2 + \pi^2}} = 0.6$ Dominant pole, $s_d = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2} = -0.6 \times 12 \pm j12\sqrt{1-0.6^2}$ P does not lie on root locus and hence simple gain adjustment is not possible
2	ANGLE TO BE CONTRIBUTED BY LEAD NETWORK $\varphi = \phi_n - 180$ Net angle contribution by poles and zeros of uncompensated system to $\phi_n = \sum \phi_p - \sum \phi_z = \angle_0^{-7.2+j9.6} + \angle_{-8}^{-7.2+j9.6} = 126.9 + 85.2 = 212^\circ$ $\varphi = 212.1 - 180 = 32^\circ$ Note: $\phi_{zc} - \phi_{pc} = 32^\circ$
3	POLE & ZERO OF LEAD COMPENSATOR

	Transfer function, $G_c(s) = \frac{K}{s+16}$
4	OPEN LOOP TRANSFER FUNCTION OF COMPENSATED SYSTEM $G_o(s) = \frac{K(s+9)}{s(s+8)(s+16)}$ Magnitude criterion, $G_o(s) _{s=-7.2+j9.6} = 1$ $\frac{K(-7.2+j9.6+9)}{(-7.2+j9.6)(-7.2+j9.6+8)(-7.2+j9.6+16)} = 0.00649K \angle 180^\circ$ $0.00649K = 1$ $K = 154$ $G_o(s) = \frac{154(s+9)}{s(s+8)(s+16)}$
5	ERROR REQUIREMENT $K_v = \lim_{s \rightarrow 0} s G_o(s) = \frac{154 \times 9}{8 \times 16} = 10.8$ K_v is greater than 10.

NOTE:

Net angle contribution of uncompensated system at desired pole location $s = -7.2 \pm j9.6$

$$G(s) \Big|_{s=-7.2+j9.6} = \frac{K}{(-7.2+j9.6)(-7.2+j9.6+8)} = 0.009K \angle 148^\circ$$

Angle contribution of lead network at desired pole location $s = -7.2 \pm j9.6$,

$$G_c(s) \Big|_{s=-7.2+j9.6} = \frac{(-7.2+j9.6+9)}{(-7.2+j9.6+16)} = 0.75 \angle 32^\circ$$

Net angle contribution of compensated system at desired pole location $s = -7.2 \pm j9.6$

$$G_o(s) \Big|_{s=-7.2+j9.6} = \frac{K(-7.2+j9.6+9)}{(-7.2+j9.6)(-7.2+j9.6+8)(-7.2+j9.6+16)} = 0.0065K \angle 148^\circ$$

	$q = 0 \Rightarrow \phi_A = \pm 180^\circ$ $q = 1 \Rightarrow \phi_A = \pm 180^\circ$ $q = 2 \Rightarrow \phi_A = \pm 300^\circ$ (same as 60°)
5	The centroid (intersection of asymptotes) is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n - m} = \frac{0 - 4 - 7}{3} = -3.67$
6	Breakaway points $\frac{dK}{ds} = 0$ $s^3 + 11s^2 + 28s + K = 0$ $K = -s^3 - 11s^2 - 28s$ $\frac{dK}{ds} = -3s^2 - 22s - 28 = 0$ $3s^2 + 22s + 28 = 0$ $s = -1.64 \text{ or } -5.7 \text{ (But, -5.7 does not lie on root locus)}$ $s = -1.64$
7	$s^3 + 11s^2 + 28s + K = 0$ Substitute $s = j\omega$, $-j\omega^3 - 11\omega^2 + j28\omega + K = 0$ Equating real and imaginary parts, $-\omega^3 + 28\omega = 0 \text{ and } -11\omega^2 + K = 0$ $\omega = \pm 5.3 \text{ and } K = 308$

1 DOMINANT POLE:

$$M_p = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}$$

$$\frac{-\xi\pi}{\sqrt{1-\xi^2}} = \ln(M_p) = -2.07$$

$$\xi\pi = 2.07 \times \sqrt{1-\xi^2}$$

$$(\xi\pi)^2 = 2.07^2 \times (1-\xi^2)$$

$$\xi = \frac{2.07}{\sqrt{2.07^2 + \pi^2}} = 0.55$$

$$\text{Dominant pole, } s_d = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2} = -0.55 \times 8 \pm j8\sqrt{1-0.55^2}$$

2 ANGLE TO BE CONTRIBUTED BY LEAD NETWORK

$$\varphi = \phi_n - 180$$

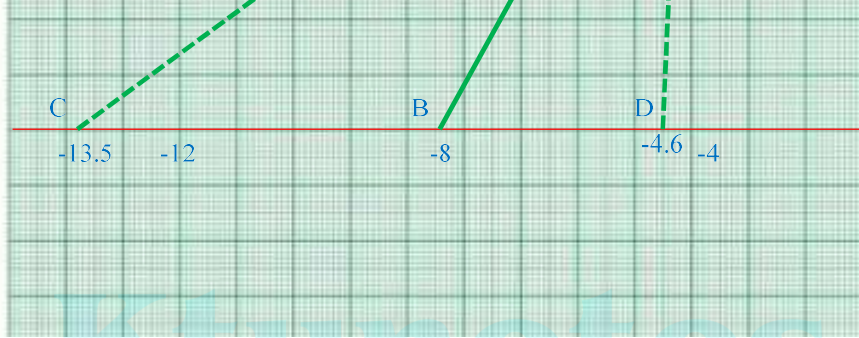
Net angle contribution by poles and zeros of uncompensated system to

$$\phi_n = \sum \phi_p - \sum \phi_z = \angle_0^{-4.4+j6.7} + \angle_{-4}^{-4.4+j6.7} + \angle_{-7}^{-4.4+j6.7} = 123.3 + 93.4$$

$$\varphi = 286 - 180 = 106^\circ$$

Since the angle contribution is more than 60° , the lead compensator is two compensators with each compensator contributing half of the required angle.

	$P_c = -\frac{1}{\alpha T} = -13.5$ $\alpha = 0.34$ Transfer function, $G_c(s) = \frac{(s+4.6)^2}{(s+13.5)^2}$ (Two compensators in cascade)
4	OPEN LOOP TRANSFER FUNCTION OF COMPENSATED SYSTEM $G_o(s) = \frac{K(s+4.6)^2}{s(s+4)(s+7)(s+13.5)^2}$ Magnitude criterion, $G_o(s) _{-7.2+j9.6} = 1$ $G_o(s) = \frac{K(-4.4+j6.7+4.6)^2}{(-4.4+j6.7)(-4.4+j6.7+4)(-4.4+j6.7+7)(-4.4+j6.7+13.5)}$ $= 0.00091K \angle 180^\circ$ $0.00091K = 1$ $K = 1099$ $G_o(s) = \frac{1099(s+4.6)^2}{s(s+4)(s+7)(s+13.5)^2}$
5	ERROR REQUIREMENT $K_v = \lim_{s \rightarrow 0} s G_o(s) = \frac{1099 \times 4.6^2}{4 \times 7 \times 13.5^2} = 4.56$ K_v is greater than 2.5.



Note: See the zero of the lead compensator is directly below the desired root loc

Consider a unity feedback system with open loop transfer function, $G(s)$ compensator to meet the following specifications i) $\%M_p \leq 35\%$ ii) settling time < 4 s iii) acceleration error constant $K_a > 2$

1 DOMINANT POLE:

$$M_p = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}$$

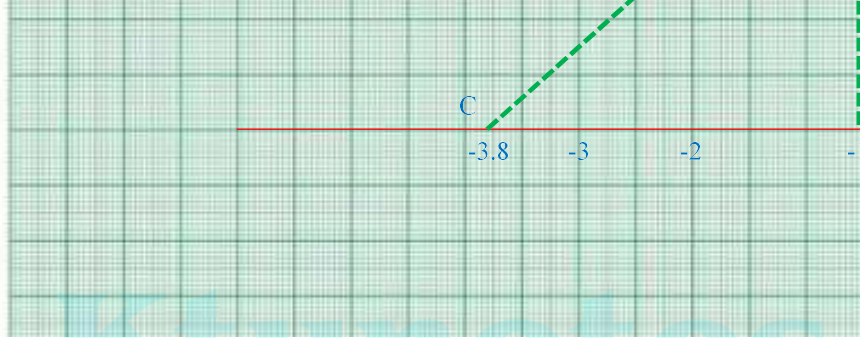
$$\frac{-\xi\pi}{\sqrt{1-\xi^2}} = \ln(M_p) = -1.05$$

$$\xi\pi = 1.05 \times \sqrt{1-\xi^2}$$

$$(\xi\pi)^2 = 1.05^2 \times (1-\xi^2)$$

$$\xi = \frac{1.05}{\sqrt{1.05^2 + \pi^2}} = 0.32$$

	Place the zero of the compensator directly below the desired location a. Draw line PC at $\varphi = \angle 47^\circ$ Point C gives pole of the compensator. Pole of compensator $P_c = -3.8$ Zero of compensator $Z_c = -1$ $Z_c = -\frac{1}{T} = -1$ $T = 1$ $P_c = -\frac{1}{\alpha T} = -3.8$ $\alpha = 0.26$ Transfer function, $G_c(s) = \frac{(s+1)}{(s+3.8)}$
4	OPEN LOOP TRANSFER FUNCTION OF COMPENSATED SYSTEM $G_o(s) = \frac{K(s+1)}{s^2(s+3.8)}$ Magnitude criterion, $G_o(s) _{-1+j2.5} = 1$ $ G_o(s) = \left \frac{K(-1+j2.5+1)}{(-1+j2.5)^2(-1+j2.5+3.8)} \right = 0.0758K$ $0.0758K = 1$ $K = 13.2$ $G_o(s) = \frac{13.2(s+1)}{s^2(s+3.8)}$
5	ERROR REQUIREMENT $K_a = \lim_{s \rightarrow 0} s^2 G_o(s) = \frac{13.2 \times 1}{3.8} = 3.47$ K_a is greater than 2.



LAG COMPENSATOR

RULES FOR DESIGN OF LAG COMPENSATOR USING ROOT LOCUS	
1	List the system specifications and translate them into a desired root locus.
2	Sketch the uncompensated root locus, and determine whether the desired root locus can be realized with an uncompensated system (with gain adjustment only).
3	If a compensator is necessary, calculate the loop gain at the desired root locus to determine the system error constant.
4	Compare the uncompensated error constant with the desired error constant. Determine the necessary increase that must result from the pole-zero ratio β of the compensator.
5	With the known ratio of the pole-zero combination of the compensator, determine the location of the pole and zero of the compensator so that the compensator root locus pass through the desired root location. Locate the pole and zero near the origin in comparison to ω_n .

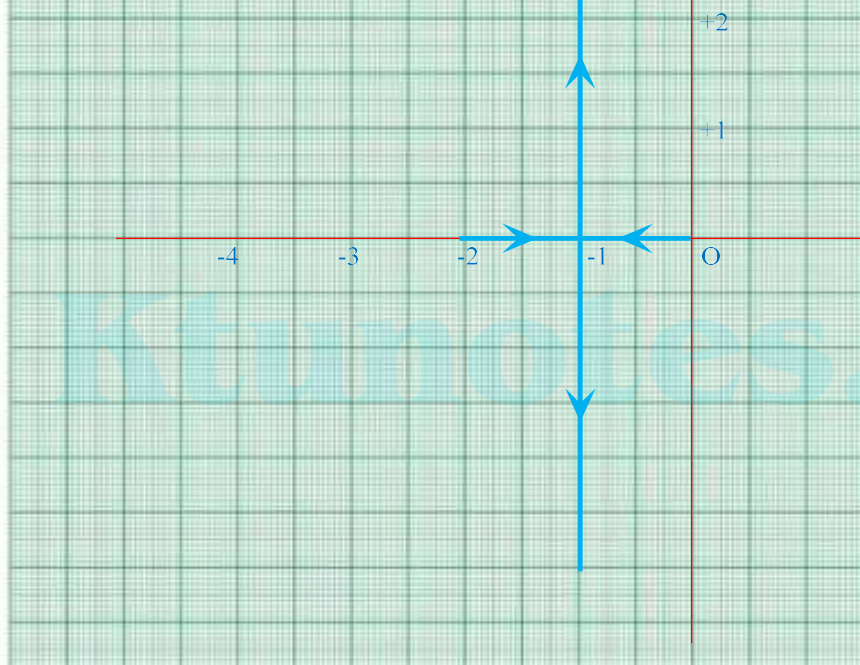
	$q = 0 \Rightarrow \phi_A = \pm 180^\circ$ $q = 1 \Rightarrow \phi_A = \pm 180^\circ$ $q = 2 \Rightarrow \phi_A = \pm 300^\circ$ (same as 60°)
5	The centroid (intersection of asymptotes) is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n - m} = \frac{0 - 1 - 3}{3} = -1.33$
6	Breakaway points $\frac{dK}{ds} = 0$ $s^3 + 4s^2 + 3s + K = 0$ $K = -s^3 - 4s^2 - 3s$ $\frac{dK}{ds} = -3s^2 - 8s - 3 = 0$ $3s^2 + 8s + 3 = 0$ $s = -0.45 \text{ or } -2.2 \text{ (But, -2.2 does not lie on root locus)}$ $s = -0.45$
7	Intersection with imaginary axis $s^3 + 4s^2 + 3s + K = 0$ Substitute $s = j\omega$, $-j\omega^3 - 4\omega^2 + j3\omega + K = 0$ Equating real and imaginary parts, $-\omega^3 + 3\omega = 0 \text{ and } -4\omega^2 + K = 0$ $\omega = \pm j1.73 \text{ and } K = 12$

1	DOMINANT POLE: $t_s = \frac{4}{\xi \omega_n}$ $\omega_n = \frac{4}{0.5 \times 10} = 0.8 \text{ rad/sec}$ Dominant pole, $s_d = -\xi \omega_n \pm j \omega_n \sqrt{1 - \xi^2} = -0.5 \times 0.8 \pm j 0.8 \sqrt{1 - 0.5^2}$
2	GAIN K AT DOMINANT POLE $ G(s) = \left \frac{K}{s(s+1)(s+3)} \right _{s=-0.4+j0.693} = 1$ $\left \frac{K}{(-0.4+j0.693)(-0.4+j0.693+1)(-0.4+j0.693+3)} \right _{s=-0.4+j0.693} = 1$ $K = 1.973$
3	DETERMINATION OF β $G(s) = \frac{1.973}{s(s+1)(s+3)}$ $K_v = \lim_{s \rightarrow 0} s G(s) = \frac{1.973}{1 \times 3} = 0.657$ Desired $K_v = 5.5$

	βT
5	TRANSFER FUNCTION OF COMPENSATED SYSTEM Transfer function of lag compensator, $G_o(s) = \frac{1.973(s+0.1)}{s(s+1)(s+3)(s+0.01)}$
6	ERROR REQUIREMENT $K_v = \lim_{s \rightarrow 0} s G_o(s) = \frac{1.973 \times 0.1}{1 \times 3 \times 0.01} = 6.58$ K_v is greater than 5.5. The design is acceptable.

Design a lag compensator for a system with open loop transfer function, $G(s)$ the following specifications i) damping ratio 0.45 ii) settling time = 4sec iii) $K_v > 20$

1	Number of branches of root locus = order of the characteristic polynomial
2	Branches of the root locus originates from open loop poles at $s = 0$ and at infinity.
3	On the real axis, the root locus is found between section $s = 0$ and $s = \infty$
4	2 branches proceed to infinity along the asymptotes $\phi_A = \frac{\pm 180(2q+1)}{n-m} \quad q = 0, 1$ $q = 0 \Rightarrow \phi_A = \pm 90^\circ$ $q = 1 \Rightarrow \phi_A = \pm 270^\circ$
5	The centroid (intersection of asymptotes) is given by $\sigma_A = \frac{\text{sum of poles} - \text{sum of zeros}}{n-m} = \frac{0-2}{2} = -1$
6	Breakaway points $\frac{dK}{ds} = 0$



1	DOMINANT POLE: $t_s = \frac{4}{\xi \omega_n}$ $\omega_n = \frac{4}{0.45 \times 4} = 2.22 \text{ rad/sec}$ Dominant pole, $s_d = -\xi \omega_n \pm j \omega_n \sqrt{1 - \xi^2} = -0.45 \times 2.22 \pm j 2.22 \sqrt{1 - 0.45^2}$
2	GAIN K AT DOMINANT POLE $ G(s) = \left \frac{K}{s(s+2)} \right _{s=-1+j2} = 1$

	The pole and zero of the compensator are placed very close to the origin. The pole is at $s = -2$. Place the zero of the compensator at $s = -0.2$. The second open loop pole is at $s = 0.1 \times -2 = -0.2$. Zero of compensator $z_c = \frac{1}{T} = -0.2$ Pole of compensator $p_c = \frac{1}{\beta T} = \frac{-0.2}{10} = -0.02$ Transfer function of lag compensator, $G_c(s) = \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} = \frac{s + 0.2}{s + 0.02}$
5	TRANSFER FUNCTION OF COMPENSATED SYSTEM Transfer function of lag compensator, $G_o(s) = \frac{5(s + 0.2)}{s(s + 2)(s + 0.02)}$
6	ERROR REQUIREMENT $K_v = \lim_{s \rightarrow 0} s G_o(s) = \frac{5 \times 0.2}{2 \times 0.02} = 25$ K_v is greater than 25. The design is acceptable.

LEAD-LAG COMPENSATOR

RULES FOR DESIGN OF LEAD-LAG COMPENSATOR USING ROOT LOCUS	
1	List the system specifications and translate them into a desired root locus.
2	Sketch the uncompensated root locus, and determine whether the desired root location can be realized with an uncompensated system (with gain adjustment only).
3	If a compensator is necessary, place the zero of the phase-lead network at the desired root location or to the left of the first two real poles.
4	Determine the pole location so that the total angle at the desired root location is zero; therefore it is on the compensated root locus.

Design a lead-lag compensator for a system with open loop transfer function, satisfy the following specifications i) damping ratio 0.5 ii) undamped oscillations $\omega_n = 5 \text{ rad/s}$ iii) velocity error constant $K_v > 80 \text{ sec}^{-1}$

1	DOMINANT POLE: Dominant pole, $s_d = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2} = -0.5 \times 5 \pm j5\sqrt{1-0.5^2} =$ P does not lie on root locus and hence simple gain adjustment is not possible
2	ANGLE TO BE CONTRIBUTED BY LEAD NETWORK $\varphi = \phi_n - 180$ Net angle contribution by poles and zeros of uncompensated system to ϕ_n $\phi_n = \sum \phi_p - \sum \phi_z = \angle_0^{-2.5+j4.3} + \angle_{-0.5}^{-2.5+j4.3} = 120.2 + 115 = 235.2^\circ$ $\varphi = 235.2 - 180 = 55.2^\circ$
3	POLE & ZERO OF LEAD COMPENSATOR Draw horizontal line PA Draw line PO Draw bisector PB so that $\angle APB = \angle BPO = \frac{\angle APO}{2}$ Draw PC and PD so that $\angle CPB = \angle BPD = \frac{\phi}{2} = \frac{55.2^\circ}{2} = 27.6^\circ$ Pole of compensator $P_c = -10$ Zero of compensator $Z_c = -2.6$ $Z_c = -\frac{1}{T} = -2.6$ $T = 0.385$ $P_c = -\frac{1}{\alpha T} = -10$ $\alpha = 0.26$ Transfer function, $G_c(s) = \frac{s+2.6}{s+10}$

	K_v is insufficient.
6	DETERMINATION OF β $K_v = 24.6$ Desired $K_v = 80$ Factor by which K_v is to be increased = $A = \frac{80}{24.6} = 3.25$ $\beta = 1.2 \times A = 1.2 \times 3.25 = 4$
7	TRANSFER FUNCTION OF LAG COMPENSATOR The pole and zero of the compensator are placed very close to the origin. The pole is placed at $s = -0.5$. Place the zero of the compensator very close to the origin. The zero is placed at $s = 0.1 \times -0.5 = -0.05$. Zero of compensator $z_c = \frac{1}{T} = -0.05$ Pole of compensator $p_c = \frac{1}{\beta T} = \frac{-0.05}{4} = -0.0125$ Transfer function of lag compensator, $G_c(s) = \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} = \frac{s + 0.05}{s + 0.0125}$
8	TRANSFER FUNCTION OF COMPENSATED SYSTEM Transfer function of lag compensator, $G_o(s) = \frac{5(s + 2.6)(s + 0.05)}{s(s + 0.5)(s + 10)(s + 0.0125)}$
9	ERROR REQUIREMENT $K_v = \lim_{s \rightarrow 0} s G_o(s) = \frac{47.4 \times 2.6 \times 0.05}{0.5 \times 10 \times 0.0125} = 98.6$ K_v is greater than 80. The design is acceptable.

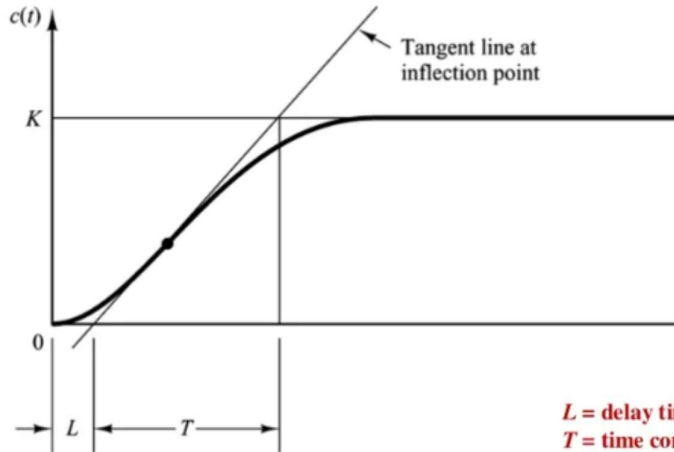
ZIEGLER-NICHOLS OPEN LOOP TUNING METHOD

Ziegler and Nichols suggested rules for tuning PID controllers (to determine the gain K_p , integral time T_i and derivative time T_d) based on the experimental step response of the plant.

First obtain experimentally the response of the plant to a unit step input as shown in the figure below.



The first tuning method is applied for plants that show an S-shaped characteristic. The S-shaped curve is characterized by two constants—the delay time, L and time constant, T . A tangent line is drawn to the S-shaped curve at the point of inflection (where the slope of the curve is maximum). The delay time, L is determined by the intersection of the tangent line with the time axis. The time constant, T is determined by intersection of tangent line with final value $c(t)=K$.



Ziegler and Nichols suggested setting the values of the parameters K_p , T_i , and T_d based on the following Table.

PID controller has a pole and origin and double zeros at $s = -1/L$.

Features of Ziegler Nichols open loop tuning:

1. applicable to stable object with no oscillations
2. easy to use
3. some processes do not permit step response tests or it gives a little process, the step input applied should be small enough for the response to be of linearity

Ktunotes.